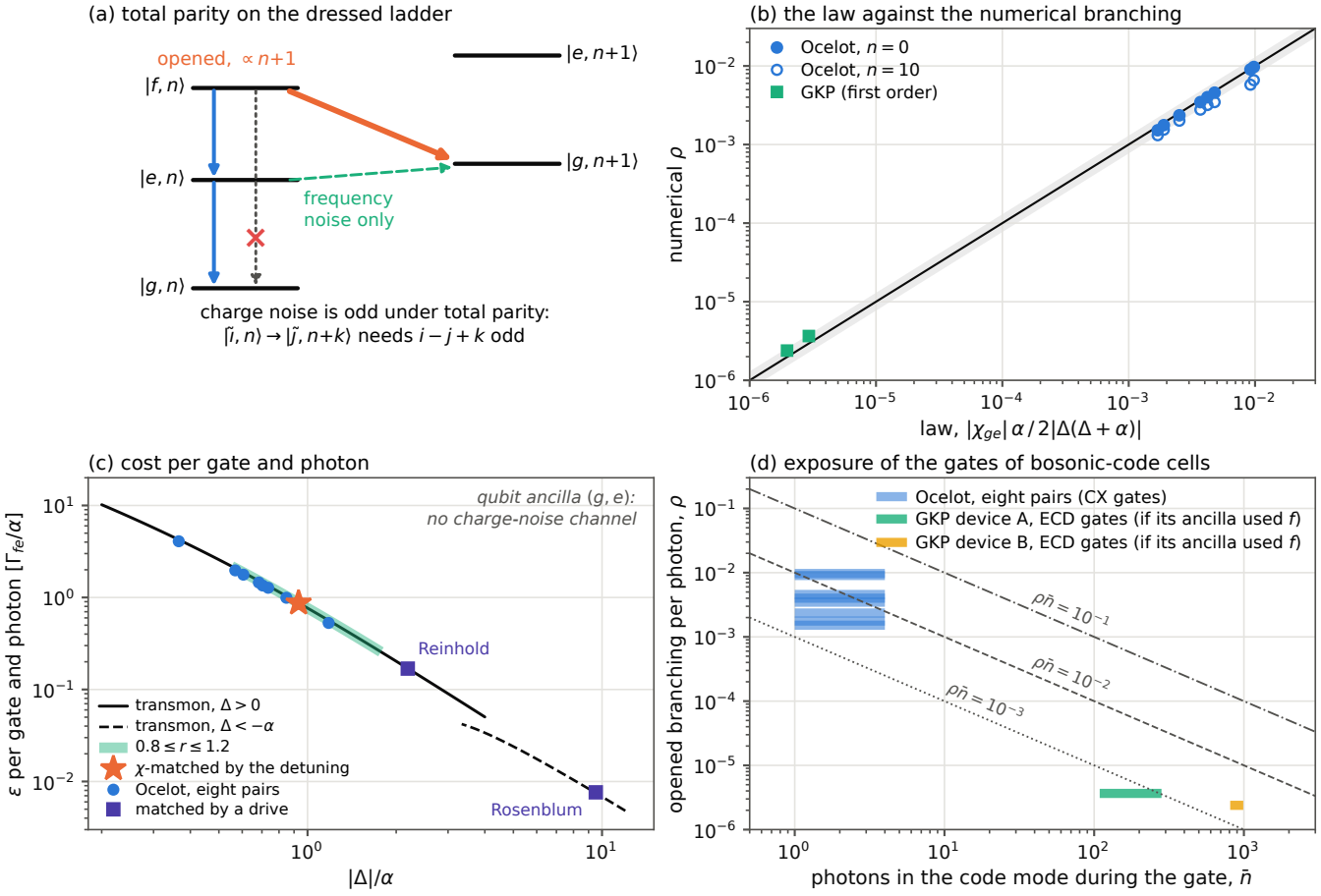
(a) total parity on the dressed ladder
opened, $\propto n+1$
$|f,n\rangle$
$|e,n+1\rangle$
$|e,n\rangle$
$|g,n+1\rangle$
frequency
noise only
$|g,n\rangle$
charge noise is odd under total parity:
$|i,n\rangle \to |j,n+k\rangle$ needs $i-j+k$ odd
(b) the law against the numerical branching
Ocelot, $n=0$
Ocelot, $n=10$
GKP (first order)
numerical ρ
law, $|\chi_{ge}|\,\alpha/2|\Delta(\Delta+\alpha)|$
(c) cost per gate and photon
qubit ancilla (g,e):
no charge-noise channel
Reinhold
Rosenblum
transmon, $\Delta>0$
transmon, $\Delta<-\alpha$
$0.8\le r\le 1.2$
χ-matched by the detuning
Ocelot, eight pairs
matched by a drive
ε per gate and photon $[\Gamma_{fe}/\alpha]$
$|\Delta|/\alpha$
(d) exposure of the gates of bosonic-code cells
Ocelot, eight pairs (CX gates)
GKP device A, ECD gates (if its ancilla used f)
GKP device B, ECD gates (if its ancilla used f)
$\rho\bar{n}=10^{-1}$
$\rho\bar{n}=10^{-2}$
$\rho\bar{n}=10^{-3}$
opened branching per photon, ρ
photons in the code mode during the gate, $\bar{n}$